Preliminary Data-Centric Study of Pulmonary Electrical Impedance Tomography with Atlas-Aware Continuous Conductivity Reconstruction

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Abstract

Pulmonary electrical impedance tomography (EIT) differs from benchmark phantom reconstruction because the underlying conductivity fields are dominated by stable thoracic anatomy and only a relatively small portion of the image varies across samples. This paper presents a preliminary pulmonary EIT study based on synthetic thorax data generated within the KTC2023 codebase. We compare FCUNet-style image-space reconstruction, direct low-frequency DCT prediction, and several atlas-aware conductivity regressors. The experiments show that pulmonary conductivity prediction is strongly atlas-dominated: on a 2048-sample synthetic test split, a train-mean atlas baseline already achieves a masked relative-L_2 error of 0.2552, while a direct continuous DCT regressor degrades to 0.3509. Atlas-aware residual, hybrid, and atlas-decoder models recover the correct data regime but remain effectively tied to the atlas baseline globally and inside residual regions. In parallel, pulmonary segmentation on the same synthetic setting shows that FCUNet currently outperforms the DCT route in accuracy, whereas DCT remains substantially more efficient in parameter count and inference latency. Finally, a first matched 16-electrode time-difference study shows that local change structure can be recovered better than a zero predictor in active regions, but false positives still prevent global RMSE gains. These findings suggest that the key pulmonary bottleneck is no longer global image representation but measurement-to-residual inference. They also indicate that future pulmonary work should move toward 16-electrode matched simulation, residual or time-difference targets, and local deviation modeling rather than pure absolute-conductivity prediction.

Keywords: pulmonary EIT, conductivity regression, atlas prior, low-frequency reconstruction, residual imaging

1 Introduction

Pulmonary EIT is attractive for bedside monitoring because it is radiation-free, portable, and capable of measuring regional impedance changes in real time [2]. At the same time, it is more structurally constrained than generic phantom reconstruction: the global thoracic outline is stable, whereas clinically meaningful change is often concentrated in relatively small local deviations such as regional ventilation loss, collapse, or fluid-related contrast.

This makes pulmonary EIT a poor fit for a naive interpretation of image reconstruction as unconstrained pixel prediction. A method that appears strong on benchmark phantoms may still fail on pulmonary data if it cannot separate stable global anatomy from measurement-driven local perturbations. The goal of this preliminary paper is therefore not to claim a final pulmonary solution, but to document what the current repository already reveals about pulmonary representation bias.

At the same time, the exploratory branch should not remain closed within only custom architectures. Pulmonary EIT already has a widely used classical reconstruction route, namely GREIT [1], and it also has reusable Python tooling through pyEIT [4]. Recent pulmonary deep-learning work such as Zeng et al. [5] further suggests that lung tdEIT benefits from

representation-aware learning under matched measurement protocols. For that reason, the current pulmonary study is now being repositioned as a two-stage process: first reproduce externally grounded pulmonary baselines, and then adapt the repository's own DCT / residual ideas on top of them.

2 Study Setting

2.1 Synthetic Pulmonary Data

The repository currently contains synthetic thorax-style datasets with two lungs, a heart-like region, and configurable local pathology perturbations. The main datasets used in this paper are:

- `dataset_lung_2k`: 2048 samples in the current 32-electrode, 2356-channel KTC-style measurement format;
- `dataset_lung_varlow`: 512 samples with weaker anatomical and local perturbation complexity;
- `dataset_lung_varhigh`: 512 samples with stronger anatomy variation, pathology variation, and conductivity texture.

These data support both a segmentation-style evaluation and a continuous conductivity-regression evaluation.

2.2 Real Thoracic Measurements

The repository also includes real thoracic `.get` files from a PLOS One dataset related to lung-heart separation in EIT [3]. We verified that these recordings follow a 16-electrode acquisition layout and reduce to 208 valid channels after DCT-EIT-style reordering. This is important because it means the current 32-electrode neural pipelines are not yet directly matched to the real pulmonary measurement regime.

2.3 Methods

The main pulmonary methods examined so far are:

- FCUNet segmentation baseline;
- DCT predictor for pulmonary segmentation;
- direct DCT conductivity regression;
- atlas-residual DCT conductivity regression;
- hybrid atlas + coarse DCT + local refinement regression;
- atlas-decoder regression with a learned atlas-aware residual decoder;
- FC-SigmaUNet as a heavier continuous image-space baseline.

3 Pulmonary Segmentation Results

On the current 2048-sample synthetic pulmonary segmentation task, FCUNet remains the stronger baseline:

- FCUNet mean score: 0.7819,
- DCT predictor mean score: 0.7272,
- DCT predictor ensemble mean score: 0.7269.

However, the DCT route remains much more efficient:

- DCT predictor: 3.85M parameters, 2.14 ms single-sample latency, 0.25 ms/sample at batch=32;
- FCUNet: 40.70M parameters, 28.82 ms single-sample latency, 14.60 ms/sample at batch=32.

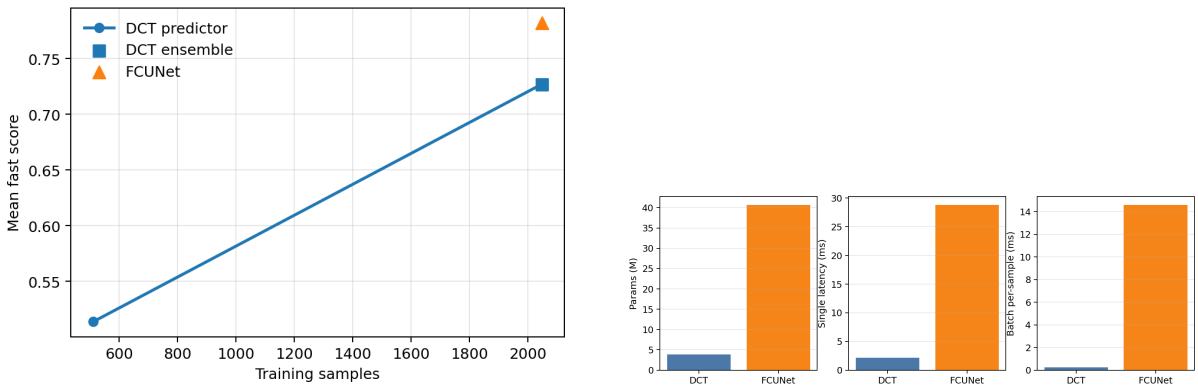

Figure 1: Pulmonary segmentation summary. Left: DCT data-scaling behavior. Right: DCT versus FCUNet efficiency comparison.

Therefore, the pulmonary segmentation evidence currently supports an efficiency-oriented role for DCT rather than a best-accuracy claim.

4 Continuous Conductivity Regression

4.1 Atlas Dominance

The most important conductivity-regression result is that pulmonary synthetic data are strongly atlas-dominated. On `dataset_lung_2k`, a trivial train-mean atlas baseline already achieves:

$$\text{RelL2} \approx 0.2552.$$

By contrast, the first direct continuous DCT regressor degrades to:

$$\text{RelL2} \approx 0.3509.$$

So the pulmonary problem is not primarily "recover the whole thorax from scratch", but rather "recover small local deviations around a stable anatomical template".

4.2 Atlas-Aware Adaptation

To match this structure, we implemented three atlas-aware models:

- atlas-residual DCT,
- hybrid DCT with coarse and local branches,
- atlas-decoder with a learned residual image decoder.

On `dataset_lung_2k`, all three return almost exactly to the atlas baseline:

- atlas-residual DCT: 0.2552,
- hybrid DCT: 0.2552,
- atlas-decoder: 0.2552.

This means atlas-aware adaptation prevents collapse, but does not yet yield a meaningful improvement beyond the atlas itself.

4.3 Complexity-Matched Evidence

The same conclusion also holds on matched low/high-complexity pulmonary datasets. In low complexity:

- atlas baseline: 0.1990,
- direct DCT: 0.6944,
- atlas-residual DCT: 0.2008,
- hybrid DCT: 0.2012,
- atlas-decoder: 0.2006,
- FC-SigmaUNet pilot: 0.2049.

In high complexity:

- atlas baseline: 0.2995,
- direct DCT: 0.7230,
- atlas-residual DCT: 0.2971,
- hybrid DCT: 0.2973,
- atlas-decoder: 0.2972,
- FC-SigmaUNet pilot: 0.3010.

These experiments indicate that the core pulmonary challenge is not decoder capacity. Even a much heavier FC-SigmaUNet pilot stays near the same atlas-dominated operating point.

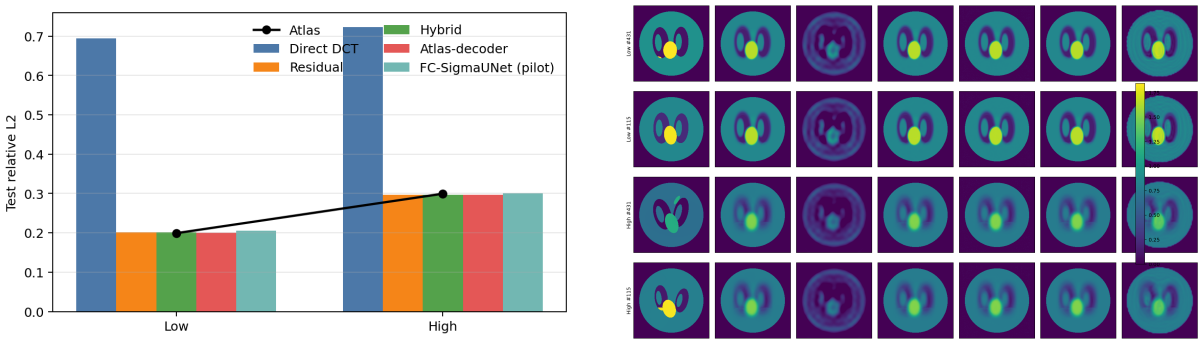

Figure 2: Matched low/high-complexity pulmonary conductivity results. Left: relative-L_2 summary. Right: qualitative comparison.

5 Residual-Focused Analysis

Global relative-L_2 can still hide whether a model actually improves inside the deviation regions that matter clinically. We therefore evaluated only pixels satisfying

$$|\sigma - \sigma_{\text{atlas}}| > 0.08.$$

On `dataset_lung_2k`, the residual-region relative-L_2 values are:

- atlas: 0.3746,
- atlas-residual DCT: 0.3747,
- hybrid DCT: 0.3747,
- atlas-decoder: 0.3746.

So the atlas-aware models do not merely *appear* tied to the atlas globally; they are also essentially tied to the atlas in the local deviation regions themselves.

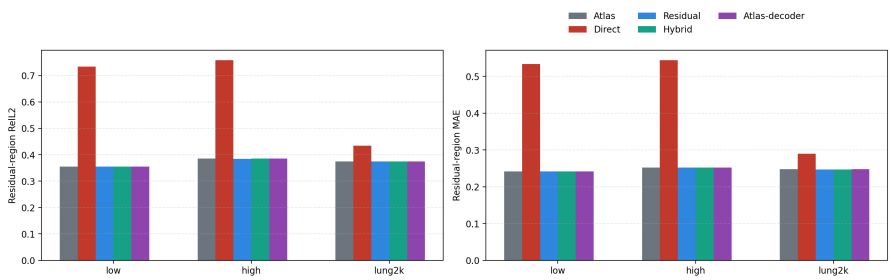

Figure 3: Residual-focused pulmonary analysis using deviation-region metrics.

We also checked whether a focus-weighted residual loss was the missing ingredient. It was not. Removing focus weighting changes low-complexity relative-L_2 from 0.2008 to 0.2007 and high-complexity relative-L_2 from 0.2971 to 0.2972. Hence simple loss reweighting is not the real bottleneck.

6 Matched 16-Electrode Time-Difference Exploration

The real thoracic recordings in this repository are not only pulmonary in content; they are also acquired under a different measurement regime from the 32-electrode KTC-style synthetic chain. To reduce that mismatch, I implemented a first matched 16-electrode simulation and learning pipeline:

- paired pulmonary phantoms with a reference state and a target state,
- raw 256-channel measurement simulation,
- DCT-EIT-style 256 → 208 channel reordering,
- direct prediction of continuous conductivity change $\Delta\sigma$ from reordered delta measurements.

This new line should be interpreted as an exploratory bridge toward realistic pulmonary EIT rather than a final model.

6.1 Classical GREIT Reproduction

To avoid evaluating the TD16 branch only against internal neural baselines, I also reproduced a classical GREIT baseline using pyEIT [4] and a custom `Draeger 208` protocol aligned to the local `256 -> 208` channel reordering. This is important because GREIT is still the most standard pulmonary linear reconstruction family in practice [1], and because the real thoracic `.get` data in this repository follow the same `16`-electrode measurement regime.

On the mixed TD16 pilot, the reproduced GREIT baseline reaches:

- thorax mesh: $\mathrm{RMSE}_{32} = 0.0859$, active-region relative-$L_2 = 1.0034$,
- circle mesh: $\mathrm{RMSE}_{32} = 0.0851$, active-region relative-$L_2 = 0.9951$.

On the active-only TD16 pilot:

- thorax mesh: $\mathrm{RMSE}_{32} = 0.1205$, active-region relative-$L_2 = 0.9910$,
- circle mesh: $\mathrm{RMSE}_{32} = 0.1279$, active-region relative-$L_2 = 1.0203$.

These numbers are not yet strong pulmonary results, but they matter for two reasons. First, they show that a classical pulmonary baseline can now be reproduced under the same local TD16 measurement protocol rather than only discussed abstractly. Second, they indicate that the current learned TD16 models are already competing with, and in some settings surpassing, a matched GREIT baseline even though they still do not outperform the zero predictor globally on mixed data.

6.2 Approximate VAE-Manifold Reproduction

To move beyond classical baselines, I also implemented an approximate reproduction of the manifold-learning route described by Zeng et al. [5]. The local approximation keeps the same high-level idea: first train a variational autoencoder on TD16 conductivity-change images, then train a multilayer perceptron to map TD16 measurements into the learned latent space, and finally decode the latent prediction back to conductivity-change images. The exact CT-derived 15000-pair training pipeline of the original paper is not yet available locally, so the present implementation should be treated as a matched-protocol approximation rather than a faithful end-to-end reproduction.

On the active-only TD16 pilot, the VAE autoencoder itself achieves RMSE = 0.1189 and active-region relative-$L_2 = 0.7587$, while the latent predictor degrades to RMSE = 0.1315 and active-region relative-$L_2 = 0.9692$. This indicates that, under active-only conditions, the dominant bottleneck is the inverse mapping from measurements to latent variables rather than the image manifold. On the mixed TD16 pilot, however, the story changes: the VAE autoencoder already collapses near the weak-change operating point (RMSE = 0.0856, active-region relative-$L_2 = 0.9983$), and the latent predictor remains essentially tied to that weak manifold (RMSE

$= 0.0842$, active-region relative-$L_2 = 0.9983$). Therefore, the manifold-learning route is plausible only when the data regime is already change-dominant; once zero-change samples dominate, the learned manifold itself becomes too insensitive to sparse local changes.

I further tested a two-stage combination in which a mixed-data external mask-only localizer gates the output of an active-only VAE predictor. This reduces the mixed-pilot RMSE from 0.0891 to 0.0862, but the active-region relative-L_2 worsens from 0.9525 to 0.9753. So stronger localization can indeed suppress false positives for the VAE route, yet it still does not solve the actual amplitude recovery problem in the active region.

I then made the coupling tighter by training a mask-conditioned latent predictor in which the mixed-data predictor explicitly regresses a low-frequency spatial mask and feeds that mask representation into the latent head. This reduces the mixed-pilot active-region relative-L_2 to 0.9686, but the global RMSE remains 0.0886, which is still worse than both the mixed VAE predictor (0.0842) and the simpler external soft-fusion strategy (0.0862). Therefore, the failure is not simply that the latent predictor never sees localization cues. The remaining problem is that localization and residual amplitude recovery are still not coupled in a sufficiently effective way.

6.3 Mixed Pilot with Zero-Change Cases

On a 64-sample pilot dataset containing both active-change and near-zero-change pairs, the zero predictor remains a strong global baseline:

- zero baseline: RMSE $= 0.0835$, active-region relative-$L_2 = 1.0000$,
- TD16 predictor: RMSE $= 0.0958$, active-region relative-$L_2 = 0.9788$.

The zero-delta fraction is 42.9%. This means the learned model begins to recover some local change structure better than the trivial baseline, but false positives still hurt the global RMSE.

6.4 Active-Only Pilot

When the zero-change proportion is reduced, the same pattern becomes clearer:

- zero baseline: RMSE $= 0.1314$, active-region relative-$L_2 = 1.0000$,
- TD16 predictor: RMSE $= 0.1321$, active-region relative-$L_2 = 0.9228$.

So once the task focuses on actual pulmonary change, the model starts to beat the zero predictor in local change reconstruction, but it still does not surpass it on global RMSE.

6.5 Sparse TD16 Follow-Up

To suppress diffuse false positives, I added inactive-region upweighting and a weak prediction L_1 sparsity term. This improves local change recovery further, but again at the cost of worse global RMSE:

- mixed sparse TD16: RMSE $= 0.1038$, active-region relative-$L_2 = 0.9255$,
- active-only sparse TD16: RMSE $= 0.1353$, active-region relative-$L_2 = 0.9129$.

I then added a sample-level change-aware gate that predicts whether a meaningful conductivity deviation exists and scales the residual output accordingly. On the mixed pilot, this gate improves the trade-off relative to sparse-only TD16:

- gate only: RMSE $= 0.1012$, active-region relative-$L_2 = 0.9521$,
- gate + sparse: RMSE $= 0.1007$, active-region relative-$L_2 = 0.9525$.

I then replaced the sample-level gate with a spatial change-mask branch. This model predicts both low-frequency conductivity change and a low-frequency binary change mask, and multiplies the two. On the mixed pilot, this becomes the best learned TD16 variant so far:

- spatial-mask TD16: RMSE = 0.0923, active-region relative-L_2 = 0.9620.

This is better than the original mixed TD16 (‘0.0958‘, ‘0.9788‘) and better than the sample-gated variants in global RMSE. On the active-only pilot, the same spatial-mask design reaches:

- RMSE = 0.13136,
- active-region relative-L_2 = 0.9463.

This almost exactly matches, and very slightly improves upon, the active-only zero baseline RMSE (‘0.13139‘) while still outperforming the zero predictor in active-region reconstruction. Therefore explicit spatial change localization is currently the most promising pulmonary TD16 direction in this repository.

I also tested active-sample oversampling on the mixed pilot. It does not change the conclusion: the oversampled spatial-mask model reaches RMSE = 0.0925 and active-region relative-L_2 = 0.9608, which is effectively tied to the non-oversampled spatial-mask model. So the remaining bottleneck is not merely sample imbalance.

However, even the best mixed-data spatial-mask variant still does not beat the mixed-data zero predictor on global RMSE (‘0.0835‘). Therefore the next pulmonary step should not be a larger absolute-image decoder. It should be a more explicit change-aware model that jointly addresses:

- change/no-change detection,
- local residual reconstruction,
- false-positive suppression in near-zero regions.

I tested one such follow-up directly by conditioning the residual coefficient branch on the predicted low-frequency spatial mask coefficients. This conditional variant also supports mask teacher forcing during training. The result is informative but not yet decisive. On the mixed pilot, conditional residual regression reaches RMSE = 0.0950 and active-region relative-L_2 = 0.9504, which is still worse than the simpler spatial-mask model (‘0.0923‘, ‘0.9620‘). Full teacher forcing does not improve the result. On the active-only pilot, the conditional model slightly lowers global RMSE to 0.13131, marginally below the spatial-mask model (‘0.13136‘) and the zero baseline (‘0.13139‘), but it still does not beat the original active-only TD16 model in local change recovery. Hence the remaining issue is no longer just whether spatial change should be modeled explicitly; it is how to couple change localization with residual amplitude estimation more effectively.

To sharpen that diagnosis, I added an oracle-mask analysis. The trained model is kept fixed, but the predicted spatial mask is replaced at evaluation time by the ground-truth change mask. For the conditional model, I also tested an even stronger oracle in which the residual branch is conditioned on the ground-truth low-frequency mask coefficients. This experiment reveals the current bottleneck very clearly. On the mixed pilot, the spatial-mask TD16 model improves from RMSE = 0.0923 to 0.0771 under oracle masking, and the conditional model improves further to 0.0739 when both oracle conditioning and oracle masking are provided. On the active-only pilot, the corresponding best oracle result is RMSE = 0.1137, again far below the learned standard-output result (‘0.1313‘). Therefore the residual branch is already capable of reconstructing local pulmonary change better than the raw benchmark numbers imply. The dominant remaining TD16 error now appears to be spatial change localization rather than residual amplitude regression itself.

To test that diagnosis more directly, I then trained a dedicated mask-only TD16 predictor whose sole target is the binary spatial change region $1(|\Delta\sigma| > 0.02)$. This dedicated mask head indeed outperforms the internal mask branches learned jointly inside the residual models. On the mixed pilot, the external mask predictor reaches mask IoU = 0.0545, compared with 0.0285 for the spatial TD16 internal mask and 0.0359 for the conditional TD16 internal mask. On the active-only pilot, the dedicated mask head improves further to mask IoU = 0.0898, again above the internal spatial and conditional mask branches (‘0.0436‘ and ‘0.0535‘ respectively). This confirms that localization quality can be improved by specializing the objective.

However, a second result is equally important: simply taking this better external mask and multiplying it with the existing residual predictions does not produce a strong global reconstruction gain. On the mixed pilot, external soft fusion with the best spatial residual model yields RMSE = 0.0926, essentially tied with the original spatial TD16 result (‘0.0923‘). On the active-only pilot, the same external-mask fusion again produces an almost exact tie in global RMSE. So the updated pulmonary TD16 conclusion is slightly more specific than before. The bottleneck is not merely “mask quality” in isolation. It is the coupling between spatial change localization and residual amplitude regression. Future pulmonary TD16 work should therefore pursue more tightly coupled two-stage or jointly conditioned localization-plus-residual models, rather than a simple post-hoc mask replacement strategy.

7 Discussion

The preliminary pulmonary evidence leads to four clear conclusions.

First, pulmonary EIT should not be treated as a generic full-image generation problem. The strong mean-atlas baseline shows that a large fraction of the task is already explained by stable anatomy.

Second, direct absolute-conductivity prediction is structurally unstable. The direct DCT regressor collapses, and atlas-aware reformulations are necessary even to stay in the correct pulmonary image regime.

Third, the unresolved difficulty is now sharply localized: the main problem is not how to decode a pulmonary image once a good representation is known, but how to infer the local residual from the measurement vector.

Fourth, the current pulmonary line is still only partially matched to the available real data. The original synthetic training chain remains 32-electrode and KTC-style, whereas the real thoracic data in this repository are 16-electrode and 208-channel. The first TD16 study now confirms that matched simulation is technically feasible and scientifically informative, but it also shows that change detection and false-positive control must become first-class design targets. A serious pulmonary continuation should therefore move toward:

- a 16-electrode matched simulation pipeline,
- residual imaging rather than pure absolute conductivity prediction,
- time-difference targets when sequential pulmonary measurements are available.

8 Conclusion

This preliminary pulmonary paper does not yet claim a final pulmonary EIT solution. Instead, it establishes the current state of the repository in a clear way:

- FCUNet is the stronger pulmonary segmentation baseline at the current scale;
- DCT remains attractive because of its efficiency and clear data-scaling behavior;

- continuous pulmonary conductivity regression is strongly atlas-dominated;
- atlas-aware models stabilize reconstruction but do not yet recover local deviations beyond atlas-level quality;
- future pulmonary work should shift toward 16-electrode matched simulation and residual or time-difference reconstruction targets.

This makes the pulmonary branch mature enough for a preliminary write-up and also clarifies the next technically justified research direction.

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